MIL-S-21520C(Wep)
7 June 1962
Superseding
MIL-S-21520B(Wep)
28 April l961

MILITARY SPECIFICATION
SEALS, INTERTUBE
FOR
LAUNCHER TUBE
MARK 4 MODS 0,1,2, AND 3 AND
MARK 8 MODS 0 and 1

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - This specification states the requirements for the intertube seals for Launcher Tube, Mark 4 Mods 0, 1, 2, and 3 and Mark 8 Mods 0 and 1.

2. APPLICABLE DOCUMENTS

2.1 Documents - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-C-843	Cushioning Material, Cellulosic
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof--

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Cent)

MILITARY

MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-L-21260	Lubricating Oil, internal Combustion Engine, Preservative
MIL-F-21568	Fluids, Silicone, Dimethyl Polysiloxane

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

BuWep Dwg 2024829	Seal, Inter-Tube, Lower
BuWep Dwg 2025022	Seal, Inter-Tube, Upper
SA-2440203	Endurance Test Fixture
SA-2619377	Lower Seal Test Fixture
SA-2619378	Upper Seal Test Fixture

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

NAVWEPS OP 400	General Instructions for Design, Manufacture, and Inspection of Naval Ordnance Equipment
NAVORD INST 4355.1	Equipment for Use by Government inspector in Final Inspection of Naval Ordnance Items
BUWEPS INST 4355.5	Bureau of Naval Weapons Calibration Program
Standards Laboratory Infomation Manual	

(Copies of specifications, standards, drawings, and publications required ‘by contractions in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications - The following document form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description - The intertube seals are ring-shaped devices constructed of nylon fabric impregnated with a Buna Neoprene compound. The seals are used for isolating the space between two nominally concentric tubes from external conditions of sea water and air pressure.

3.2 General Requirements

3.2.1 Preproduction Sample - Unless otherwise specified in the contract or Order, preproduction samples of the upper and lower seals shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification (refer to 4.2.1). At the discretion of the procuring agency, the requirement for a preproduction sample may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

3.2.2 Materials - Unless otherwise specified in contract, equisition, or order, all materials and purchased parts entering into the final assembly of the upper and lower seals shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWep Dwg 2024829 or 2025022,

3.2.3 Finishes, Coatings, Construction, and Dimensions - Finishes, coatings, constructions and dimensions for the upper and lower seals shall be in accordance with the applicable drawings, specifications, and publications listed in BuWep DWg 2024829 Or 2025022.

3.2.4 Marking - Marking of seals shall be in accordance with applicable BuWep DWg 2024829 or 2025022.

3.2.5 . Interchangeability - Individual assemblies and piece parts of assemblies fabricated in agreement with - their applicable drawings and specifications shall be interchangeable with like-numbered parts.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertiment Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptante of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Request for Deviations - The procedure for requesting deviations from this specification or any specification, publication material, or process specified herein shall be as set forth in NAVWEPS OP 400.

3.2.9 Conflicting Requirements - In event of conflict between this specification and any specification, or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances - Unless specifically noted, tolerances and accuracies specified herein are absolute.

3.3 Performance Requirements

3.3.1 Leakage - The intertube seals shall withstand a hydrostatic pressure of 195 to 205 psig and shall show no leakage.

NOTE

Leakage is a condition in which fluid accumulates in a sufficient amount to form a drop; a drop is an accumulation of fluid of sufficient size and weight to separate or drop from the surface on which it has collected. Wetting of surface at joint and at mating surface does not constitute a leak.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

3.3.2 Hydrostatic Pressure

3.3.2.1 Steady Pressure - The intertube seals shall withstand a hydrostatic pressure of 195 to 205 psig for 100 hours, shall show no leakage, and thereafter shall be capable of meeting all performance requirements of this specification except that the 100 hours shall not be in addition to the requirement specified in 3.3.2.3.

3.3.2.2 Rapidly Applied Pressure - The upper intertube seals shall withstand a hydrostatic pressure increased from 30 to 130 ± 5 psig within 0.5 to 0.7 seconds, maintained for 15 seconds, applied 25 times, shall show no leakage, and thereafter shall be capable of meeting-all performance requirements of this specification,

3.3.2.3 Pressure Cycling - The intertube seals shall withstand 50 cycles of hydrostatic pressure. A cycle shall consist of raising the pressure from 0 psig to 195 to 205 psig, maintaining the pressure for 2 hours, and then reducing the pressure to 0 psig for 2 hours. After cycling the seals shall show no damage and shall be capable of meeting all performance requirements of this specification except the 50 cycles shall not be in addition to the requirement specified in 3.3.2.1.

3.3.3 Flexure - The intertube seals shall withstand 300 cycles of flexure at the rate of 1 cycle-per-minute and thereafter shall be capable of meeting all performance requirements of this specification. A cycle of flexure shall consist of moving the inner circumference of the seal, relative to the periphery through an 8 by 12 inch elliptical path with the major axis of the ellipse perpendicular to the diameter of the seal.

3.4 Environmental Conditions

3.4.1 High Temperature - The intertube seals shall withstand exposure to air at a temperature of +110 ± 4 degrees Fahrenheit with a relative humidity of 95 to 98 percent for 12 hours, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

3.4.2 Low Temperature - The intertube seals shall withstand with stand exposure to air at a temperature of -20 ± 4 degrees Fahrenheit for 24 hours, shall show no damages and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.3 Thermal Shock - The intertube seals shall withstand temperature change from +80 ± 3 degrees to +28 ± 3 degrees Fahrenheit within 5 seconds, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.4 Submersion - The intertube seals shall withstand submersion in sea water at temperatures between +28 and +100 degrees Fahrenheit for 90 consecutive days, shall show no damage, and thereafter shall be capable of meeting all performance requirements of this specification.

3.4.5 Exposure to Hydraulic Fluid - Convex surfaces of the upper and both surfaces of the lower intertube seals shall withstand exposure to silicone fluid (Specification MIL-F-21568) at a temperature of +65 ± 20 degrees Fahrenheit for 90 consecutive days, shall show no damage, and thereafter shall be capable or meeting all performance requirements of this specification.

3,4.6 Eject Temperature/Pressure - The lower intertube seals shall be tested for ability to withstand temperature and pressure changes associated with operation of the Mark 7 Mod O Ejector Group.

3.5 Cleanup - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of all foreign matter.

3.6 Workmanship - The intertube seals shall be constructed and finished in a manner to assure compliance with all requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, arid specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in NAVWEPS OP 400. Production tests of final assemblies shall be on a 100-percent basis. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with NAVWEPS OP 400. Failed parts and components shall be submitted to the Material Review Board for disposition.

4.2.1 Preproduction Sample - The preproduction samples specified in 3.2.1 shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract or by the procuring activity for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval of the contracting activity, or the completion of tests on the preproduction Samples, shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order. Disposition of the test sample shall be at the discretion of the Material Review Board.

4.2.2 prOduction Lot - The term production lot shall mean a collection of like units of manufacture, produced in a production cycle with no change in process, formulation, or manufacturer's lot of raw material, submitted by a supplier for Government inspection. Unless otherwise specified the production lot shall not exceed 25 units,

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4.2.3 Periodic Production Sample - From each succeeding production lot of 25 or less upper seals or 25 or less lower seals, a random sample will be selected by the Government Inspector for submission to an agency assigned by the procuring activity to determine compliance of the stiple with the requirements of the contract, specifications, and drawings. Disposition of th etest sample shall be at the discretion of the Material Review Board.

4.3 Classification of Tests - The inspection and testing of the upper and lower seals shall be classified as follows:

4.3.1 Production Test - These tests are to be accomplished on each upper and lower seal being submitted for acceptance under the contract. production tests shall be either performed by the contractor and witnessed by the Government Inspectors or these tests maY be conducted bY the Government Inspector. These tests are detailed in 4.5.

4.3.11 Periodic Production Tests - Periodic production tests are those which are accomplished on samples selected as specified in 4.2.3, which are representative of the production of the seals, to determine that the production meets the requirements of this specification. These tests are detailed in 4.5 and 4.7. Acceptance shall be based on no defects in the sample. Failure of the sample will result in rejection of the production lot.

4.3.2 Preproduction Tests - Preproduction tests are those which are accomplished on samples selected as specified in 4.2.1, which are representative of the production of the upper and lower seals, to determine that the production meets the requirements of this specification. These are detailed in 4.5 and 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction approved by the procuring acttvity.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4.3.3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the upper and lower seals. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.4 Special-Test Equipment - The following items or special test equipment are required to perform the acceptance tests as set forth in the specification:

SA-2440203	Endurance Test Fixture
SA-2619377	Lower Seal Test Fixture
SA-2619378	Upper Seal Test Fixture

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall perform production tests as specified in paragraph 4.3.1 to assure that the intertube seals are in compliance with the requirements of 3.3.1.

4.5.2 Test Conditions - Unless otherwise specified, the intertube seals shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +85 degrees Fahrenheit
Altitude:	Normal, ground atmospheric pressure
Vibration:	None
Humidity:	Room ambient

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.

4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5.3.1 Calibration - Measuring equipment and instru-= mentation shall be calibrated as required by NAVORD INST 4355.1, BUWEPS INST 4355.5, and in accordance with the policies specified in the Standards Laboratory Information Manual issued by the Bureau of Naval Weapons Representative, Pomona, California.

4.5.4 Test Procedures

4.5.4.1 Materials and Dimensions - Each intertube seal shall be inspected for conformance to the requirements of paragraph 3.2.2 and 3.2.3.

4.5.4.2 Leakage Test - Each upper intertube seal mounted in test fixture SA-2619378 and each lower intertube seal mounted in test fixture SA-2619377 shall be subjected to a hydrostatic test pressure of 195 to 205 psig for 10 minutes to meet the requirements of 3.3.1.

4.6 Preproduction Tests

4.6.1 Hydrostatic Pressure Tests

4.6.1.1 Steady Pressure Test - One upper intertube seal mounted in test fixture SA-2619378 and one lower intertube seal mounted in test fixture SA-2619377 shall be subjected to a hydrostatic proof pressure of 195 to 205 psig for 100 hours to meet the requirements of 3.3.2.1.

4.6.1.2 Rapidly Applied Pressure Test - One upper intertube seal mounted in test fixture SA-2619378 shall be subjected to a hydrostatic proof pressure increased from 30 to 130 ± 5 psig within 0.5 to 0.7 seconds for 25 times to meet the requirements of 3.3.2.2.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4.6.1.3 Pressure Cycling Test - One each of the, upper and lower intertube seals (an upper seal mounted in test fixture SA-2619378 and a lower seal mounted in test fixture SA-2619377) shall be subjected to 50 pressure cycles to meet the requirements of 3.3.2.3.

4.6.2 Flexure Test - One of each type of the intertube seals, mounted in test fixture SA-2440203, shall be subjected to 300 cycles of flexure at the rate of 1 cycle-per-minute to meet the requirements specified in 3.3.3.

4.6.3 High Temperature Test - One Of each type Of the intertube seals shall be subjected to a temperature of +110 ± 4 degrees Fahrenheit with a relative humidity of between 95 and 98 percent for 12 hours to meet the requirements specified in 3.4.1.

4.6.4 Low Temperature Test - One of-each type of intertube seals shall be subjected to a temperature of -20 ± 4 degrees Fahrenheit for 24 hours to meet the requirements specified in 3.4.2.

4.6.5 Thermal Shock Test - One of each type of intertube seals shall be subjected to a temperature change from +80 ± 3 degrees Fahrenheit to +28 ± 3 degrees Fahrenheit within 5 seconds to meet the requirements specified in paragraph 3.4.3.

4.6.6 Submersion Test - To meet the requirements specified in 3.4.4, one of each type of the intertube seals shall be subjected to submersion in sea water at a temperature between +45 degrees and +100 degrees Fahrenheit for 88 consecutive days and then at a temperature 0f +28 to +32 degrees Fahrenheit for 2 days.

4.6.7 Hydraulic Fluid Test - One of each type of the intertube seals shall be exposed to hydraulic fluid (MIL-F-21568) at a temperature of +65 ± 20 degrees Fahrenheit for 90 consecutive days to meet the requirements specified in 3.4.5.

4.6.8 Eject Temperature/Pressure Test - To meet the requirement of 3.4,6, one lower intertube seal shall be subjected to five consecutive exposures to temperatures and pressures associated with operation of the Mark 7 Mod 0 Ejector Group or a prototype ejector unit having launch parameters equivalent to the Mark 7 Mod 0 Ejector Group.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

4.7 Periodic Production Test

4.7.1 Pressure Cycling Test - One upper and one lower intertube seal, selected as specified in 4.2.3, (an upper seal mounted in test fixture SA-2619378 and a lower seal mounted in test fixture SA-2619377), shall be subjected to 50 pressure cycles to meet the requirements specified in paragraph 3.3.2.3.

4.8 Rejection - Failure of the seals to meet any of the requirements and tests of this specification shall be considered. cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A - Level A packaging shall be such as to assure adequate protection to prevent corrosion, deteriorations or physical damage during shipment, handling, indeterminate storage, and world-wide distribution.

5.1.1.1 Cleaning - the seals shall be cleaned by wiping with a clean dry cloth.

5.1.1.2 Drying - Not applicable

5.1.1.3 Preservative Application - The seals shall be completely covered with a thin coat of lubricating oil, Specification MIL-L-21260, Type P-10, Grade 1, 2, or 3.

5.1.1.3.1 Bead Protection - The beads On both edges of the seals shall be covered with split-hose bead protection

5.1.1.4 Unit Packaging - The seals shall be folded with polyethylene film between contacting surfaces; wrapped with tape, Specification PPP-T-60, Type I, Class 3; and enclosed in a heat-sealed bag, Specification MIL-B-131, Class 1.

5.1.1.5 Intermediate Package - Not applicable

5.1.2 Level B - Not applicable

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

5.1.3 Level C - Level C cleaning, drying, preservation, arid unit packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Level A - Level A packing shall be such as to assure adequate protection to prevent damage during shipment, handling, indeterminate storage, and world-wide distribution.

5.2.1.1 Exterior Containers - Each seal shall be packed with cushioning material, specification PPP-C-843, in a close-fitting wooden shipping container conforming to specification PPP-B-621, Class 2, Style 4.

5.2.2 Level B - Not applicable

5.2.3 Level C - The intertube seals, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with Uniform Freight Classification Rules and other common-carrier regulations applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required by the contract or order, unit packages, intermediate packages, and exterior containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and drawing number or Intertube seals.

(c) Selection of applicable levels of preservation, packaging, and packing.

(d) Exceptions to this specification, applicable drawings, and other documents.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:47Z
Check the source to verify that this is the current version before use.

NOTICE: When Government drawings, specifications or other data are used for any prupose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.